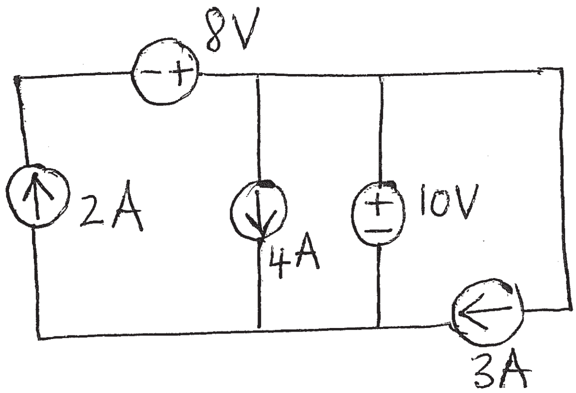
EBN111 Klastoets/*Class Test*#1: 1 Maart/*March* 2013

Q1 'n Nuwe soort battery lewer 10W drywing vir 8 ure sonder enige variasie in spanning of stroom. Na 8 ure val die drywingsuitset lineêr van 10W na 0 binne 30 minute. Bepaal die energie stoorkapasiteit van die battery. [2]

A new type of battery delivers 10W of power for 8 hours without voltage or current fluctuation. After 8 hours the power output drops linearly from 10W to 0 in 30 minutes. Determine the energy storage capacity of the battery.

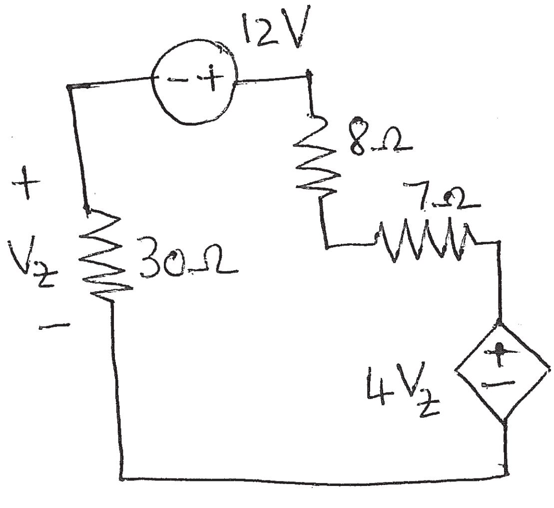
Q2 Bereken die drywing ge-assosieer met die 8V spanningsbron en die 3A stroombron. [2]

Calculate the power associated with the 8V voltage source and the 3A current source.

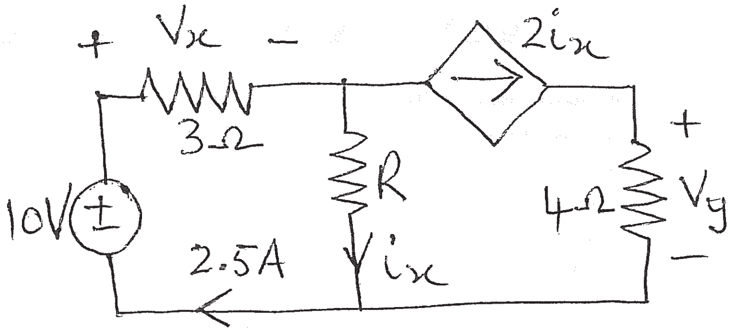


Q3 Bepaal v_z in die stroombaan. [2]

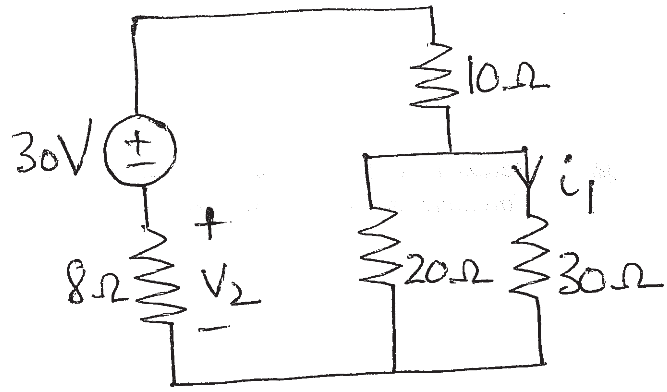
Determine v_z in the circuit.



Q4 Bereken v_x, v_y en i_x in die stroombaan. [3]
Calculate v_x, v_y and i_x in the circuit.



Q5 Bereken/*Calculate*
(a) i_1 [2]
(b) v_2 [2]



Q6 Bereken R_{eq}. [2]
Calculate R_{eq}.